Cardinal Analytx Solutions

Better Care, Sooner

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Predict Future Risk and Rising Cost

Cardinal Analytx Solutions

Significant Healthcare AI Spin-out from Stanford Start-X

4 Years
R&D

21 Million
Lives

2.5x
Accuracy

3
Products

21
Advisors and Board

28
Employees

2
Customers Paid

8
Customers MSAs

20
Pipeline Customers

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Predicting Future Risk and Rising Cost

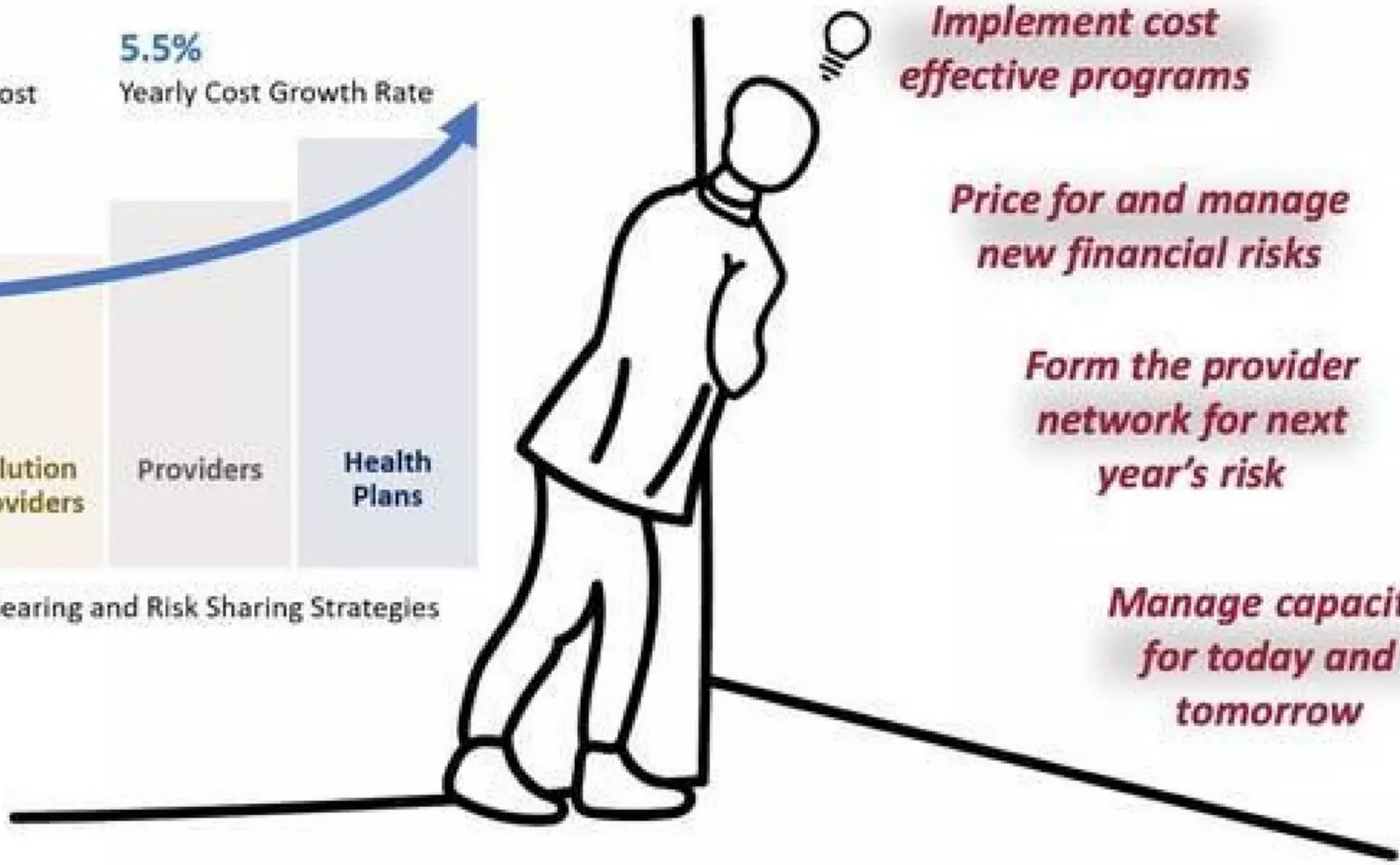
6% of today's population will account for 1/3 of next year's new high cost

$3.5 Trillion
Annual Healthcare Cost

5.5%
Yearly Cost Growth Rate



Increased Risk Bearing and Risk Sharing Strategies



Implement cost effective programs

Price for and manage new financial risks

Form the provider network for next year's risk

Manage capacity for today and tomorrow

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Predict the What, Why and When of Tomorrow's Cost

Mitigate Rising Risk

Solution 1:
COST BLOOM INTERVENTION

- Cost Bloom, Jump, High Cost Claimant Likelihood
- Engagement Likelihood
- Clinical Impactability Score
- Case Selection Optimizer
- Action Plan Recommendation

Ave Annual Price Per Customer: $850k

Steer High Cost Events

Solution 2:
STEERAGE PRECISION

- Clinical Event Likelihood
- Engagement Likelihood
- Event Timing Likelihood
- Case Selection Optimizer
- Case Summary

Ave Annual Price Per Customer: $400k

Predict Total Cost

Solution 3:
RISK ASSESSMENT

- Prospective Risk Score & Cost
- Concurrent Risk Score & Cost
- Cost Bloom, Jump, High Cost Claimant Likelihood

Ave Annual Price Per Customer: $600k

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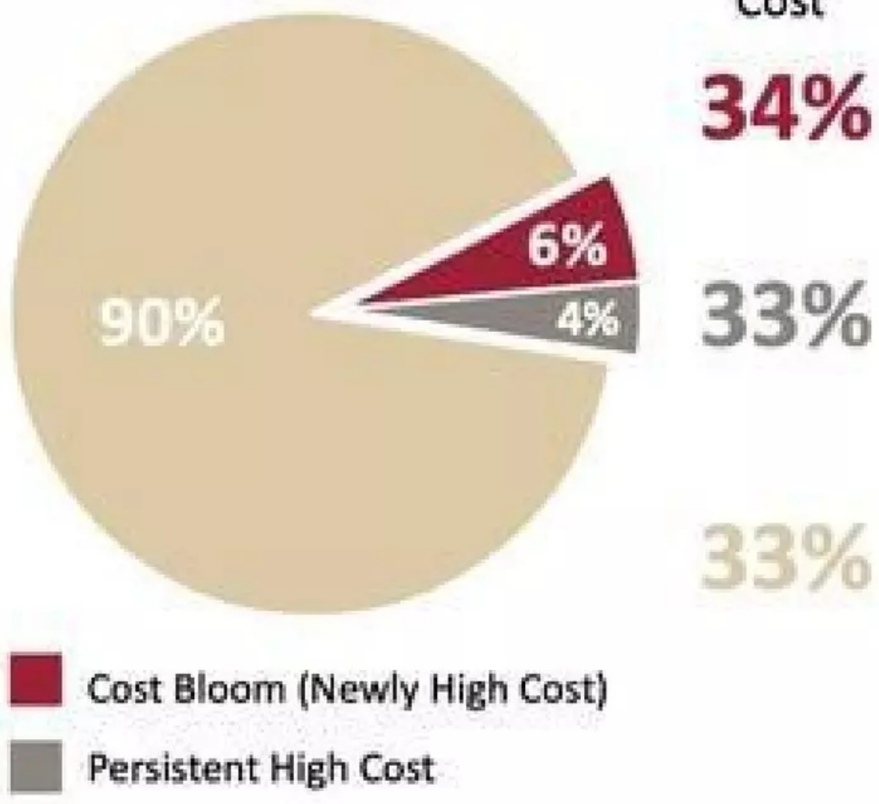
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8x ROI on Highest Impact Cost Blooms

"I felt like it was meant to be for me to reach this member, it was the perfect moment in time"
– Care Manager

Patient Outcome: 57 year old male with heart conditions, back pain, and opioid dependence was steered to a higher quality lower cost spinal fusion center, a savings of over $20k.

Health Plan A Population



Health Plan A



Cost Bloom Adoption at a Blues Plan

40% engagement rate (one call with Cost Bloom member)

26% active engagement rate (multiple calls with member)

75% successful intervention rate for actively engaged

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6x ROI on Ortho Joint Surgery Steerage

1/3
Of cost blooms were ortho joint surgeries

$3.6M savings, 6x ROI
Projected Ortho Joint savings at Health Plan A

Steerage Precision Addressing a Market Gap

Over $5M per year in Total Savings
Even larger market beyond Ortho Joint: Bariatric, High Risk OB, Hysterectomy, etc.

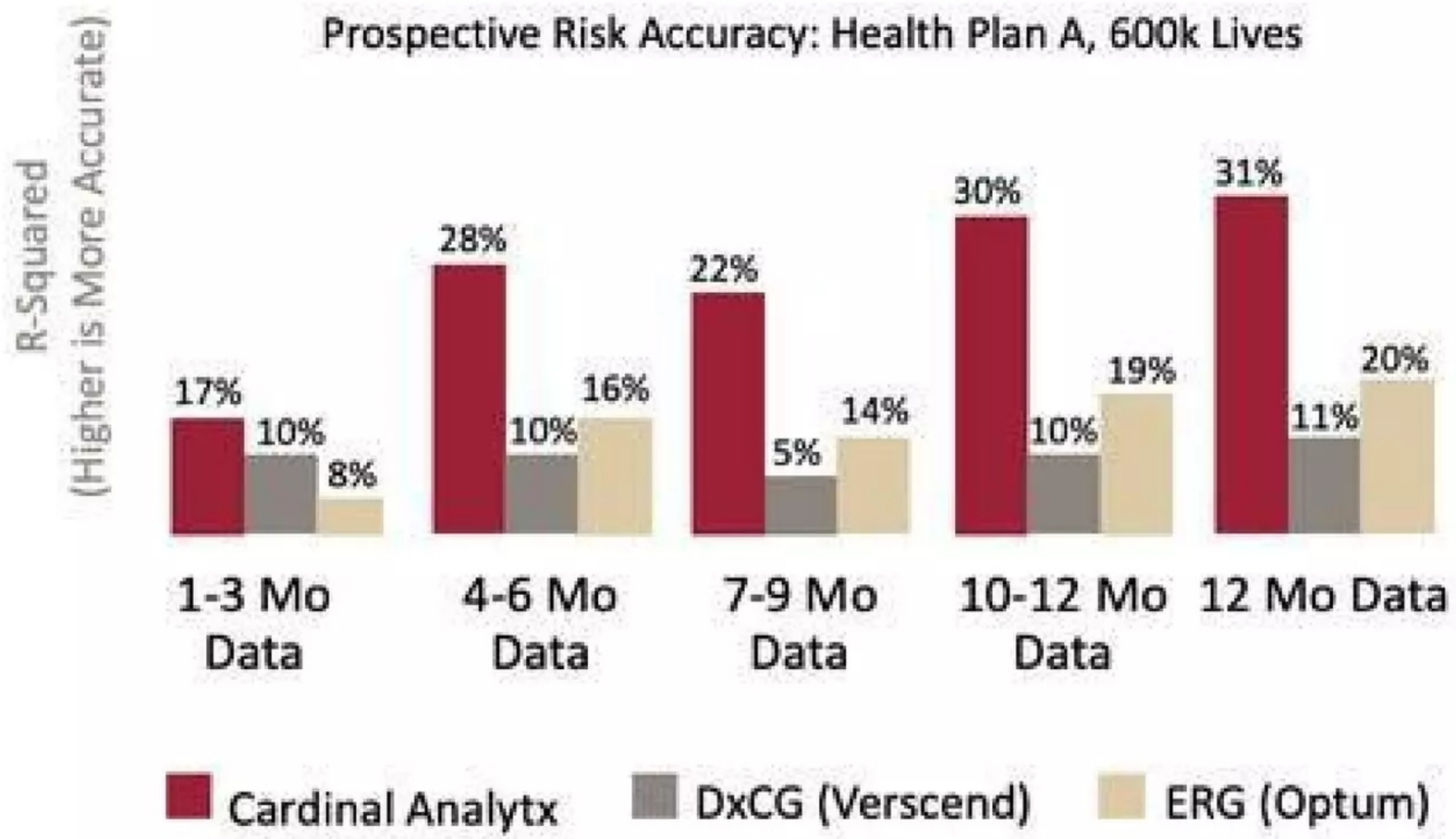
50% of Cardinal Analytx Clients Expressed Interest in Steerage
Cardinal Analytx is working with solution providers to partner (e.g., Vitals, Relay, MOBE, VIM)

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$100M Through Increased Accuracy

Our proprietary Machine Learning beats the competition



Missed Dollars by Competitors:

$63M missed dollars
Under-pricing risk

$37M missed dollars
Over-pricing / retention risk

Strong Competitive Position

Strategic Approaches	Competitive Landscape	
1. Machine Learning Predictions with Drivers and Actions	Cardinal Analytx Solutions	KenSCI (Provider) Cyft (CM platform)
2. Machine Learning Predictions + Model Correlations	Base Health (Med Adv. Risk Adj) Jvion (Provider) Ayasdi (Provider)	
3. Machine Learning Predictions	Lumiata (Risk) Cogitativo (Payment) HBI Solutions (Risk) IBM Watson (Truven)	Ark.one (Risk) NextHealth (Provider) Hindsait (Data Aggregation)
4. Traditional Predictions	Optum (Risk) Milliman (Risk) Cave Consulting (Risk) Verscend (Risk) Lexis Nexis (Risk) Deerwalk (Risk)	Health Catalyst (Provider) Welltok (Member Platform) Lightbeam (Provider, Risk) HCC (Risk) ACG (Risk)

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Professional Team



Linda Hand
CEO

- 35 years in organizational leadership, product development, solutions delivery and go-to-market strategies
- Successful exit of DecisionView to IMS Health, Clinical Tech core of IQVIA / Quintiles merger
- BA in Computer Science, UC Berkeley, Haas School of Business Executive Program



Niall O'Cathasaigh
Chief Financial Officer

- 20 years of startup experience across 15 venture backed startups, 4 Healthcare companies. 7 years as founder and fund manager. 3 years at Apple.
- CIMA Qualified. DBS from University College Dublin. BA in History from University College Galway and Universite De Poitiers.



Brian Maples
VP of Data Science

- 15 years in biomedical R&D and machine learning
- Invented and patented molecular diagnostic tech at Ionian, acquired by Alere to develop CLIA waived rapid flu test; Led Nuna's data science development of a Network Solutions product; at Stanford, created now leading ancestry inference methodologies
- PhD in Biomedical Informatics from Stanford University, MS in Statistics from Stanford University, BA in Science and Management from Claremont McKenna College



Lu Lu
Head of Product

- 12 years healthcare experience in strategic and clinical transformation initiatives focused on transition to value with over 19 Health Systems, 45 clinical departments, and 450 design participants.
- MPH from Harvard School of Public Health, BA in Psychology from UCLA.



Chris DeRienzo
Chief Medical Officer

- 11 years clinical experience, double-boarded MD
- While serving as Mission Health Chief Quality Officer gained >$65M through continuous improvement, pay for value and population health initiatives.
- Spearheaded integration of analytics and data science while reducing harm and scaling clinical pathway development.
- MD/MPP from Duke University, BS in Biology, Biochemistry, and Psychology from Brandeis University

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Engaged Founders, Investors & Board

Founders



Nigam Shah
Associate Professor of Medicine,
Stanford

- Associate Professor of Medicine, Funded by 5 NIH Institutes, FDA and 7 companies; Asst Director of Center for Biomedical Informatics Research.
- Fellow, American College of Medical Informatics and American Society for Clinical Investigation.
- MS and MD Baroda College (Baroda, India), PhD, Penn State, Postdoctoral, Medical Informatics, Stanford.



Arnold Milstein
Professor of Medicine,
Stanford

- Professor of Medicine and Director, Stanford Clinical Excellence Research Center; Elected Member Institute of Medicine Section of National Academy of Sciences; Medical Director at Pacific Business Group on Health; Board Trustee Intermountain Healthcare.
- Founder national health care performance improvement firm and Leapfrog Group; former Congressional MedPac Commissioner.
- BA, Harvard; MD, Tufts; MPH, U.C. Berkeley.

Investors

CP CARDINALPARTNERS

StartX
StanfordStartX Fund

PREMERA | BLUE CROSS

John Doerr
Family Fund

Blue Shield of California

Florida Blue

Board



Thomas McKinley
Chairman of Cardinal Analytx,
Partner at Cardinal Partners



Linda Hand
CEO of Cardinal Analytx



John Doerr
Chairman, Kleiner Perkins



Mark Smith
California HealthCare Foundation,
Co-leads a CMS effort to
accelerate value-based payment



Nigam Shah
Associate Professor of Medicine,
Stanford



Elizabeth Spaulding
Partner at Bain & Company,
Global Head of Bain Digital



Kent Marquardt
Former EVP and CFO at Premera
Blue Cross

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Industry Leading Advisors

Science



Susan Athey, PhD
Economics of Technology Professor, Stanford Graduate School of Business



Trevor Hastie
The John A. Overdeck Professor of Mathematical Sciences, Stanford University



Colt Courtright, MPA
Director, Corporate Data & Analytics

Industry



Virginia McFerran, MBA
Formerly, Partner, Optum Ventures, President and CEO, Optum Analytics



Ray Bradford, MBA
Founder and CEO, Spruce Health

Health Plan

Jeffrey Bailet, MD, MSPH
Executive VP of Health Care Quality and Affordability Blue Shield of California

David Fields
President
Dean Health Plan

John Espinola, MD, MPH,MBA
Executive VP of Healthcare Services, Premera Blue Cross

Juan Chaluja
VP Commercial Markets Analytics and Business Performance, Florida Blue

Clinical



Kevin A. Schulman, MD, MBA
Professor of Medicine, Assoc Chair, Bus Dev & Strategy, Stanford School of Medicine



Ken Paulus, MA
Past President and CEO, Allina and Atrius Health Systems



Ronald A. Paulus, MD
President and CEO, Mission Health



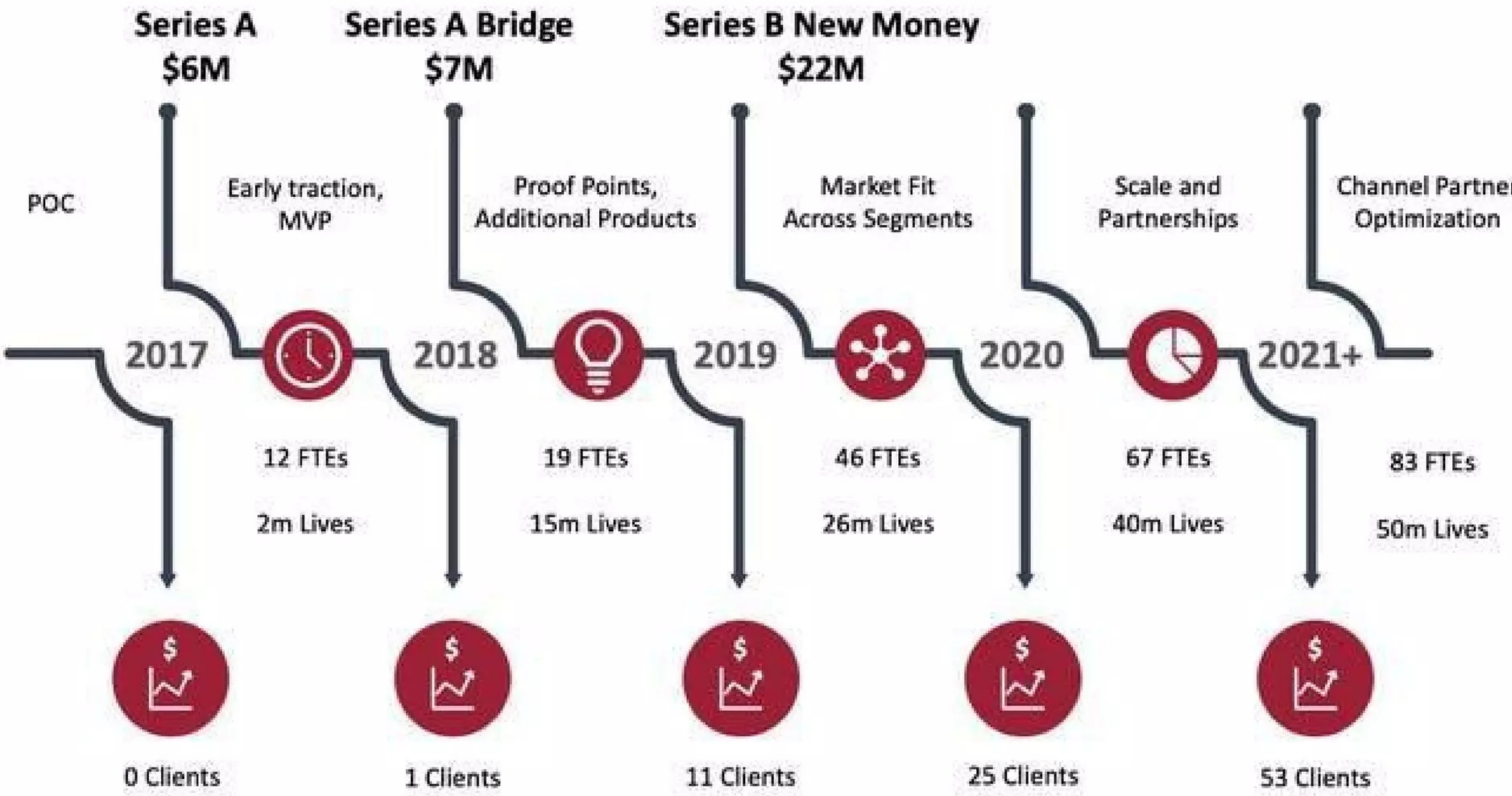
Lee Sacks, MD
Former CMO, Advocate Aurora Health



Alan Glaseroff, MD
Faculty, Stanford University's Clinical Excellence Research Center

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Investment Opportunity



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